# Self-Supervised Production Anomaly Detection and Progress Prediction Based on High-Streaming Videos

Yifan Li, Zhi-Hai Zhang, *Senior Member, IEEE*, Jiaqi Xu, Xiaowei Yue, *Senior Member, IEEE*, and Li Zheng

***Abstract*— Real-time production monitoring incorporating progress prediction and anomaly detection is essential for quality and efficiency. Traditional vision-based anomaly detection methods struggle to differentiate between production-related features and background noise, and fail to consider the heterogeneity of production stages. This paper introduces an integrated approach that merges progress prediction and anomaly detection, employing the Autoencoder Process Probability Embedding (APPE) method. APPE maps the distribution of images from normal production to a progress-related Gaussian Mixture Model (GMM), focusing on identifying production-relevant features while minimizing background interference through the proposed Spatial Activation Map (SAM). The proposed SAM improves the interpretability of the neural network by highlighting the specific features that influence the model's decisions. The method is assessed through real-world datasets in the assembly of water valves and the production of commercial aircraft spoilers. The case study shows that our approach can achieve superior effectiveness compared to the benchmark, notably improving both task performances by integrating progress prediction with anomaly detection.**

***Note to Practitioners*—In many manufacturing settings, such as aircraft production, tasks that involve human-robot collaboration or high-precision manual assembly play a significant role. The ability to detect anomalies and monitor progress in real-time is critical for ensuring the quality and efficiency of production. The manual nature of these operations makes them challenging to monitor through in-situ embedded digital sensors, yet real-time operation videos are readily available. Vision-based production monitoring has been widely used in applications such as product surface inspection, but existing algorithms often face difficulties distinguishing between normal background variations and anomalies related to production. This paper introduces a new approach, called Autoencoder Process Probability Embedding (APPE), which integrates progress recognition and anomaly detection into a cohesive monitoring task, allowing the model to differentiate between background elements and features related to production. Although our method is demonstrated in production scenarios as case studies, the proposed SAM mechanism is versatile to be applied in other contexts with similar types of categorical labels.**

***Index Terms*— Production monitoring, self-supervised learning, anomaly detection, aircraft assembly monitoring.**

## I. Introduction

PRODUCTION monitoring is vital to improve manufacturing efficiency and product quality, reduce the risk of injuries, and prevent safety accidents. Although existing methods based on sensor data show promise in automating process control [1], anomaly detection [2], production monitoring and diagnosis [3], they fall short in industrial production scenarios that involve continuous manual operations, such as human-robot collaboration tasks or manual high-precision aerospace assembly. Specifically, these methods are not sufficient for real-time tracking of progress and process surveillance because of the lack of well-targeted sensors. High-streaming videos become ubiquitous in smart factories, especially as high-resolution video surveillance systems tend to be more affordable. There is an urgent need to develop production monitoring based on real-time high-streaming videos and mine values from the collected data.

In production monitoring, there are two major problems: (i) **Progress Prediction**: Predicting the ongoing process steps; (ii)**Anomaly Detection**: Detecting abnormal activities during the production process. Vision-based production monitoring has been increasingly studied in recent years. Notably, well-designed algorithms have been developed and applied in various scenarios, including surveillance cameras [4], surface anomaly detection [5], and additive manufacturing [6]. A more comprehensive literature review will be analyzed in Section II. Although there are some research progresses, the vision-based monitoring for industrial production processes still faces unique challenges and complexity:

- Variations in unrelated pixels can significantly exceed those in areas of interest. As illustrated in Figure 1, the background variation within the green box is substantially more pronounced than the anomalous pixels in the red box. This necessitates training the model to focus more acutely on pixels pertinent to the anomaly.

Received 19 August 2024; accepted 9 November 2024. Date of publication 4 February 2025; date of current version 11 April 2025. This article was recommended for publication by Associate Editor X. Zhang and Editor K. Liu upon evaluation of the reviewers' comments. The work of Yifan Li was supported by NSFC under Grant 724B2019 and Grant 72188101. The work of Zhi-Hai Zhang was supported by NSFC under Grant 72250710683. The work of Li Zheng was supported by NSFC under Grant 72188101. The work of Xiaowei Yue was supported by Beijing National Science Foundation under Grant 3244032 and Grant L241039. *(Corresponding authors: Li Zheng; Xiaowei Yue.)*

Yifan Li, Zhi-Hai Zhang, Xiaowei Yue, and Li Zheng are with the Department of Industrial Engineering, Tsinghua University, Beijing 100190, China (e-mail: lyf21@mails.tsinghua.edu.cn; zhzhang@tsinghua.edu.cn; yuex@tsinghua.edu.cn; lzheng@mail.tsinghua.edu.cn).

Jiaqi Xu is with the Economic Policy and Development Strategy Research Center, China Waterborne Transport Research Institute, Beijing 100088, China (e-mail: xujiaqi@wti.ac.cn).

Digital Object Identifier 10.1109/TASE.2025.3538328

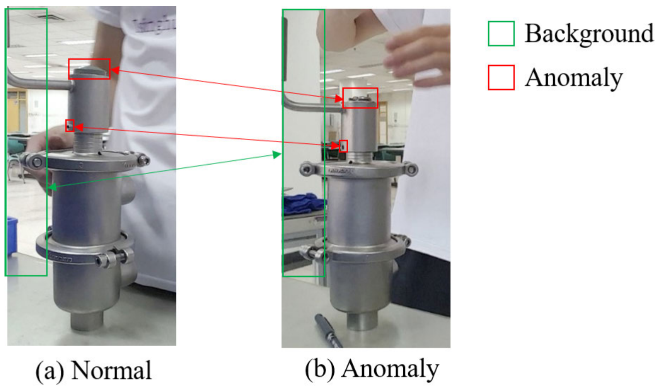


Fig. 1. An image comparison between normal and anomaly production operations.

- The appearance of workpieces undergoes considerable dynamic changes throughout the production process, resulting in spatial-temporal heterogeneity. The substantial dynamic variation and the heterogeneity make production anomaly detection extremely complicated.
- There is an inherent scarcity of abnormal data in the production process since the production line runs in a normal condition most of the time. The data scarcity and imbalance raise significant challenges for model development and parameter learning.

Due to these challenges, the typical video monitoring methods in the Computer Vision (CV) domain may not work well in industrial production monitoring. Additionally and more importantly, the anomaly classification in CV and the anomaly classification in production monitoring are quite different. We elucidate the mapping relationship between industrial production and CV in Figure 2. The first column in this figure lists the anomaly classes typically identified in the CV domain, as detailed in [4]. The second column enumerates potential anomalies that might arise during production operations. These are further consolidated in the third column, where we provide a synthesis of these anomalies along with four illustrative examples.

Anomalies within our focus encompass four categories as follows. *Foreign Object Anomalies*: These anomalies arise from the presence of extraneous or unrelated objects within the production environment, which could potentially disrupt the manufacturing process [7]. *Worker Operation Anomalies*: This category includes anomalies related to the actions or postures of workers that are discernible without reference to the workpiece. Examples include inappropriate assembly actions or unsafe behaviors [8]. *Part Interaction Anomalies*: Anomalies in this category are identifiable without observing the worker's actions. They involve the interactions between parts, such as incorrect assembly order or mispositioning, indicating a deviation from the intended assembly process [9], [10]. *Progress Correlated Anomalies*: These anomalies are specific to different assembly stages. They may involve the misuse of tools not intended for certain production stages or operations that stray from the prescribed process.

Existing research on abnormal production activities primarily focuses on predefined anomaly classes or rules [7], [11], [12]. However, the anticipation and annotation of all possible future anomalous activities become impractical, especially in dynamic production environments [4]. Consequently, there emerges a critical need for a self-supervised anomaly detection algorithm trained exclusively on normal image data. This methodology obviates the necessity for abnormal data and predefined rules, significantly augmenting the model's ability to foresee unencountered anomalies.

Moreover, traditional studies often treat progress prediction and anomaly detection as distinct, isolated tasks. It is essential to integrate anomaly detection within the predictive model to withhold stage predictions in anomalous situations. Given that anomaly detection criteria may vary across different production stages, it is imperative for a model to assimilate real-time progress information to facilitate precise and contextually relevant anomaly identification. How to integrate these two tasks into a unified model remains an open question.

This study endeavors to develop algorithms that can identify forthcoming anomalies using solely normal production image data. Utilizing the inherent characteristics of the production process as a self-supervised learning signal enables the model to concentrate on features pertinent to production, notwithstanding the extensive variation in background elements. Integrating anomaly detection with progress prediction tasks not only enhances the model's accuracy in forecasting production stages but also furnishes predictions correlated with progress-related anomalies.

The main contributions of the paper are as follows.

1) An Autoencoder Process Probability Embedding (APPE) method considering the production state information is developed to map the distribution of the normal production process.
2) A Spatial Activation Map (SAM) is proposed to train the model focus on production-related features. We provide an interpretative formulation for SAM, thereby augments the neural network's explainability and makes its operations more transparent.
3) We advocate the integration of progress prediction and anomaly prediction to enhance both tasks' performance and efficiency.

This paper is structured in the following manner: Section II offers a detailed overview of relevant research in the field. The proposed methodology is presented in Section III. This is complemented by two case studies in Section IV, which serve to demonstrate the practical effectiveness of our approach. Finally, conclusions and future work are discussed in Section V.

## II. Related Literature

As previously highlighted in Section I, the monitoring task within the assembly process encompasses two principal components: progress prediction and anomaly detection. This section offers a detailed review of vision-based techniques for both tasks. Furthermore, it delves into the attention mechanism and foreground segmentation, due to their pertinence to the proposed Spatial Activation Map (SAM).

### A. Review on Progress Prediction

Progress prediction research can be segmented into macro-scale workpiece state analysis and micro-scale action

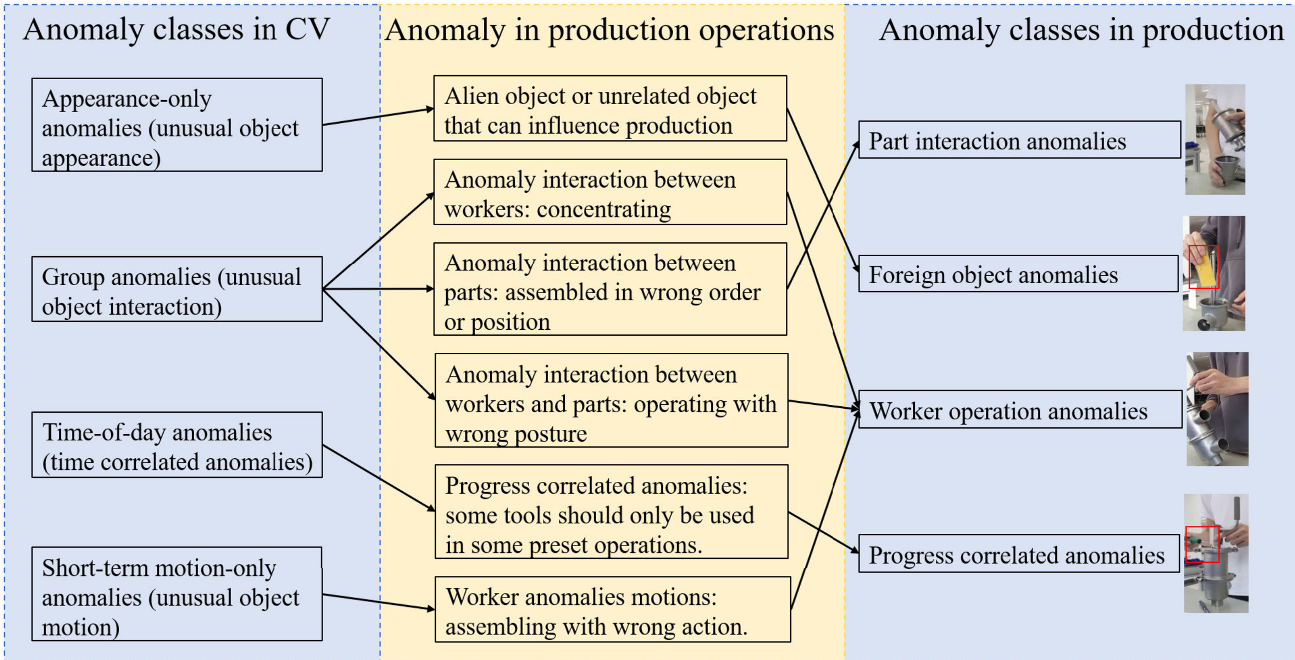


Fig. 2. Anomaly comparison between computer vision (CV) and industrial production.

recognition. On the macro-scale, the state of production can be recognized through techniques like RGB color-based pixel segmentation and edge estimation against CAD models [13], operation classification with Desnet for predefined operation classes [14], and semantic segmentation algorithms for mechanical part identification [9], [15]. Micro-scale approaches focus on identifying operators' actions through object detection frameworks [16], augmented with pose estimation [17], neural networks for part and action identification [8], and hand movement tracking to deduce picked parts [18]. However, within actual production environments, predictions may become unreliable due to anomalous situations, a critical aspect these methods often overlook.

### *B. Review on Anomaly Detection*

Anomaly detection in production processes employs a diverse array of vision-based techniques. Research in this field ranges from the application of preset rules or supervised methods for detecting anomalies to advanced neural network applications. For example, a neural network was utilized to identify assembly defects with an 87% success rate [11], whereas image-to-CAD model comparisons were implemented to ensure accurate part installation [12], and contrastive learning was employed to monitor conditions within a flotation bank [19]. Nevertheless, these approaches are generally confined to detecting anomalies defined by preset rules, a limitation in production scenarios where anomaly data may be scarce and future anomalies difficult to anticipate [4].

To address the scarcity of anomaly data and the unpredictability of anomalies, unsupervised methods have been adopted. An unsupervised, image-based process monitoring technique using deep belief networks was proposed [20], leveraging the energy of restricted Boltzmann machines as a control statistic for fire anomaly warnings. Specifically in additive manufacturing, an unsupervised method was developed to target hot-spot detection—an overheating phenomenon, and assess pixel auto-correlation [6]. A dynamic sparse subspace learning approach has been developed for online structural change-point detection of high-dimensional streaming data [21]. Furthermore, some researches [22], [23], [24] delve into surface variation monitoring and quality status evaluation in manufacturing by fusing in-plant multi-resolution measurements and process information, along with employing manifold learning for feature extraction. Within the wider context of computer vision, anomaly detection techniques are broadly categorized into reconstruction-based, probabilistic, and distance-based approaches [4]. Autoencoders (AE), renowned for their reconstruction-based strategy, are predicated on the assumption that anomalies cannot be effectively reconstructed. The Variational Autoencoder (VAE) [25] refines this methodology by using reconstruction probability as an anomaly indicator. Probabilistic approaches, like that suggested by [26], embed images within a probabilistic space, flagging an image as anomalous when its probability value dips below a predetermined threshold. On the other hand, distance-based methods develop a "normality" model from training data and evaluate deviations from this model to determine anomaly scores [27]. Nonetheless, the inability to distinguish between production-relevant pixels and background variations diminishes the efficacy of these models in dynamic environments with frequent background changes, as illustrated in Figure 1.

### *C. Review of the Attention Mechanism in Computer Vision*

The attention mechanism has gained significant popularity in the field of computer vision (CV) following the advent of transformer-based vision models [28]. Attention mechanisms exhibit remarkable diversity; for instance, self-attention [28] utilizes the input features as the *query*, *key*, and *value* simultaneously. There are also variants such as additive attention [29] and dot-product attention. However, these mechanisms primarily alter the network structure without changing the learning target and is not suitable for introducing domain preliminary knowledge. Meanwhile, a limited number of studies have explored supervised attention mechanisms, such as the work of [30], which leverages annotated arguments to establish supervised attention, thereby enabling a more focused and effective event detection process. In our case, we propose a weakly-supervised attention mechanism using the progress information to construct an attention map focusing on production-related pixels.

### D. Review on Foreground Segmentation

In our research, it is crucial for the model to discern unrelated variations in image pixels. Traditional foreground detection algorithms, such as Mixture of Gaussian (MOG), often simply categorize moving parts as the foreground. Recently, Class Activation Map (CAM) [31] has become widely used for weakly supervised identification of class-related pixels. Based on CAM, foreground attention supervised by image classes [32] is proposed to localize the object related pixels. Other research approaches use CAM as a preliminary foreground prediction, which is then refined through methods like Expectation-Maximization (EM) [33], background awareness [34], or wave function [35], [36]. While these methods are effective in localizing object-related areas, our objective differs. In our context, various types of objects may appear in the same image, and we lack object class labels, focusing instead on subsets that influence the production process. Consequently, we propose a Spatial Activation Map extracting method to determine the importance weights of pixels.

In summary, existing literature on anomaly detection and attention mechanisms within computer vision underscores a gap in addressing dynamic and complex production environments. While numerous techniques propose solutions for specific scenarios, their adaptability is often lacking in the face of rapidly changing backgrounds and the variety of object types encountered in production settings, as described in Figure 1. Moreover, the dependency on predefined rules or class labels by many approaches limits their applicability in situations devoid of such readily available information. Our proposed APPE model seeks to fill this gap by concentrating on production-relevant features without the prerequisite of explicit object class labels, thus offering a more versatile and robust solution for anomaly detection in challenging production environments. Furthermore, in contrast to existing models that segregate progress prediction and anomaly detection, our APPE model simultaneously outputs information for both anomaly detection and progress prediction. This synergistic approach not only enhances the effectiveness of both tasks but also reduces computational complexity by half (instead of training two separate models, we now get the prediction in one).

## III. Autoencoder Process Probability Embedding

In this section, we propose a probabilistic modeling network to embed image data to a Gaussian latent space for anomaly detection. The overall structure of the proposed APPE is summarized in Figure 3. Given an image $\mathbf{X} \in \mathbb{R}^{3\times w\times h}$ with height $h$ and width $w$ from a production process, the encoder maps it to low dimensional GMM distributed latent variable $\mathbf{z} \in \mathbb{R}^d$. Besides, the feature map $\mathbf{Q} \in \mathbb{R}^{l\times w'\times h'}$ with height $h'$ and width $w'$ generated just before the last convolutions block is also extracted to a Multilayer Perceptron (MLP) to generate the proposed SAM. SAM is used as attention weight to refine the encoder feature. And the decoder reconstructs the image $\mathbf{X}'$ given $\mathbf{z}$. Then the reconstruction error is calculated and weighted by SAM to generate a Spatial Activated Reconstruction Error (SARE). The sum of Negative Loglikelihood Loss ($NLL$), triplet loss and SARE is used as the loss value to train the proposed APPE model.

### A. Production Process Modeling

Distinct from conventional vision based anomaly detection, the production process exhibits heterogeneity in image features across both spatial and temporal dimensions, as depicted in Figure 4. For spatial heterogeneity, some locations of the image are production correlated while others are those background features with different importance. For temporal heterogeneity, the image features are various in different production steps. This complexity necessitates a nuanced approach to analyzing these features.

We assume that the image feature $\mathbf{z}_k \in \mathbb{R}^d$, generated at a specific progress percentage $k \in [0, 1]$, adheres to a Gaussian distribution with mean $\boldsymbol{\mu}_k \in \mathbb{R}^d$ and covariance $\boldsymbol{\Sigma}_k \in \mathbb{R}^{d\times d}$. The mean and covariance are of functional heterogeneity along the process timeline (refer to Eqs. (1)-(3)).

$$\mathbf{z}_k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \tag{1}$$

$$\boldsymbol{\mu}_k = F_\mu(k), \tag{2}$$

$$\boldsymbol{\Sigma}_k = F_\Sigma(k). \tag{3}$$

In our comprehensive study of manufacturing, certain stages of the production process, like inspection or polishing tasks, were observed to have minimal impact on the visual characteristics of the components. Based on these observations, we propose that $F_\mu(k)$ is not a function with continuous variation. Instead, it takes the form of a piece-wise constant function, demarcated by separating points $y \in \{0, \dots, n\}$, where $k_0 = 0$ and $k_n = 1$. The methodology for unsupervised determination of separable time-points $k_y$ can be defined as a change-point detection problem [37]. We further elaborate on this concept with a formula for the piece-wise function as Eq. (4) and Eq. (5).

$$F_\mu(k) = \boldsymbol{\mu}_y \quad \text{if } k_y < k \le k_{y+1}, \quad y \in \{0, \dots, n-1\}, \tag{4}$$

$$F_\Sigma(k) = \Sigma_y \quad \text{if } k_y < k \le k_{y+1}, \quad y \in \{0, \dots, n-1\}. \tag{5}$$

Consequently, the distribution of $\mathbf{z}$ aligns with a multivariate Gaussian mixture distribution, as delineated in Eq. (6):

$$p(\mathbf{z}) = \sum_{y=0}^{y=n-1} (k_{y+1} - k_y)\mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_y, \Sigma_y). \tag{6}$$

There exist two prominent categories associated with self-supervised, image-based Gaussian Mixture Model (GMM) embedding. The first category leverages Variational Autoencoders (VAEs), which optimizes the Evidence Lower Bound (ELBO) under the assumption of a GMM prior [38]. However,

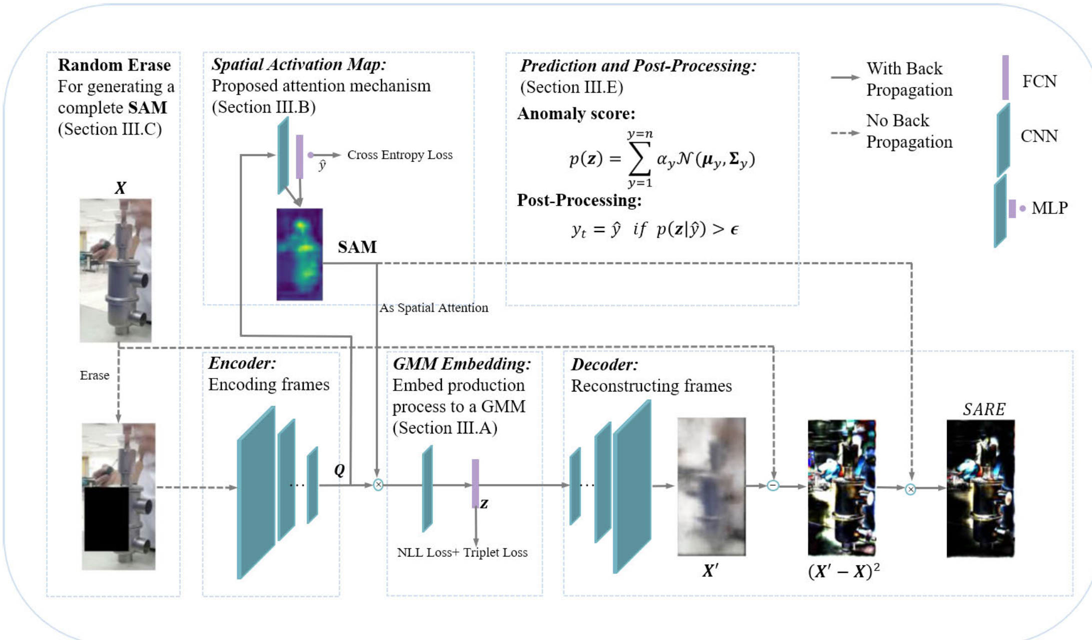


Fig. 3. The structure of the proposed APPE model.

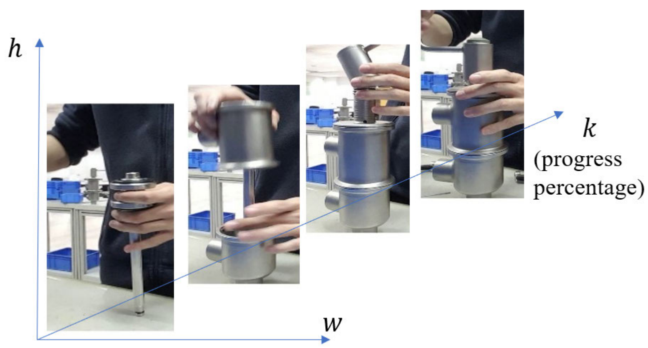


Fig. 4. Illustration of spatial-temporal heterogeneity in image features.

this approach exhibits sensitivity to outliers [39]. The second category combines Autoencoder (AE) with $NLL$ loss, aligning with other probabilistic embedding techniques [26].

In our research, we adopt the $NLL$ function as our training objective. Unlike existing methods, the cluster $y$ in our framework is predefined by the production progress $k$, which is known as priori during the training phase. Therefore, the Log-Likelihood function in our context is represented by the following equation:

$$\log p(\mathbf{z}|y) = -\frac{1}{2}\big[(\mathbf{z}-\boldsymbol{\mu}_y)^\top \Sigma_y^{-1}(\mathbf{z}-\boldsymbol{\mu}_y) + \log|\Sigma_y| + d\log(2\pi)\big]. \quad (7)$$

In this formulation, $F_\mu(k)$ and $F_\Sigma(k)$ are derived via Maximum Likelihood Estimation (MLE), providing statistical foundation for our model.

Then the training objective Negative $NLL$ can be formulated as Eq. (8).

$$NLL = \frac{1}{2}\big[(\mathbf{z}-\boldsymbol{\mu}_y)^\top \Sigma_y^{-1}(\mathbf{z}-\boldsymbol{\mu}_y) + \log|\Sigma_y|\big]. \quad (8)$$

And after every epoch of training, we update $F_\mu(k)$ and $F_\Sigma(k)$ according to maximum likelihood estimation.

### B. Spatial Activation Map for Progress Correlated Features

As shown in Figure 1, the variation of anomaly pixels can be much less than the normal background variation. Therefore, it is vital for us to employ a mechanism to let the model focus on production-related pixels. Utilizing object class as weakly-supervised signal, CAM [31] has been proposed to extract object-related pixels. However, in our cases, it is hard or even impossible to label the object classes, for each image may present hundreds of parts in a complex production process. Therefore, we propose a Spatial Activation Map ($SAM \in \mathbb{R}^{w'\times h'}$), which uses progress state $y$ as a self-supervised signal to extract a map. The idea of our approach is that when we want to recognize the proceeding process state, the weight of the pixels should be large only when it is highly relevant to the process.

To realize this intuition, we train an MLP with input feature $\mathbf{Q}$, as visualized in Figure 5. $\mathbf{Q}$ is then passed to a convolutional network with parameter $\theta$ generating a map $conv(\mathbf{Q},\theta) = \mathbf{Q}'$ with equal shape. A subsequent fully connected network (FCN) without bias is then employed to classify the process state given the feature map $\mathbf{Q}'$. We use $\mathbf{f} \in \mathbb{R}^{n\times l}$ to denote the weights matrix for FCN. The predicted class vector $\mathbf{pre} = \frac{1}{h'w'}\sum_{i=0}^{w'-1}\sum_{j=0}^{h'-1}\mathbf{f}\,\mathbf{Q}'_{:,i,j}$. And the predicted class is then calculated with a cross-entropy loss $Loss_{CE} = -log(\mathbf{pre}_y)$ to train the weights of MLP. Moreover, to emphasize features that are correlated with progress and to differentiate the embedding features among classes, we have integrated a triplet loss function, described in Eq. (9). This function aims to maintain a margin $m \in \mathbb{R}$:

$$Loss_{trip} = \max(|\mathbf{z}_{anchor} - \mathbf{z}_{positive}|^2 - |\mathbf{z}_{anchor} - \mathbf{z}_{negative}|^2 + m^2, 0). \quad (9)$$

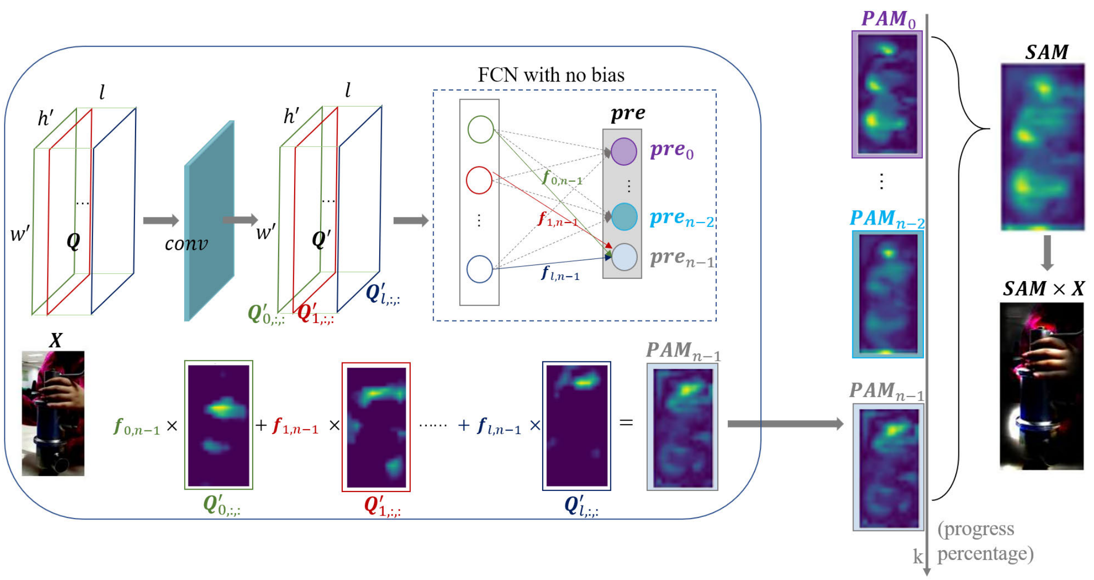


Fig. 5. The visualized procedure for producing SAM.

In this equation, $\mathbf{z}_{positive} \in \mathbb{R}^d$ refers to an encoded feature vector $\mathbf{z}$ from the same progress class as $\mathbf{z}_{anchor} \in \mathbb{R}^d$, whereas $\mathbf{z}_{negative} \in \mathbb{R}^d$ originates from a different progress class. During training, we randomly extract two images of the same class to act as the anchor and positive images, alongside an image from a different class designated as the negative, and encode them to get these three feature vectors.

Before computing the SAM, we initially derive a Progress Activation Map (**PAM**) that maps the significance of each progress class. In the absence of global pooling, we obtain the following:

$$\mathbf{PAM}_{y,i,j} = abs(\mathbf{f}_{:,y}^{\top}\mathbf{Q}'_{:,i,j}). \tag{10}$$

We use $abs(\cdot)$ to calculate **PAM** because we aim to measure the magnitude of influence a spatial location has on the production state class, regardless of whether the influence is positive or negative. Within this framework, $\mathbf{PAM}_{y,i,j}$ represents the progress scores at each spatial location, offering a precise quantification of the impact that the location $(i, j)$ has on predicting the outcome as class $y$. This metric essentially mirrors the significance of the specific spatial feature's contribution to the overall result. A negligible absolute value of this score implies that discarding the image feature at this location would not substantially influence the current stage of progress information. This indicates that the feature at this specific location holds limited relevance to the process stage $y$. Consequently, we can compute SAM according to Eq. 11, which signifies the spatial feature importance in the production process. And this interpretative formulation augments the neural network's explainability and making its operations more transparent.

$$SAM = \frac{1}{n}\sum_{y=0}^{n-1}\mathbf{PAM}_y. \tag{11}$$

With this interpretation of SAM, we propose to use it as an attention weight both in the encoder and decoder. To be specific, in the encoder, we multiply SAM with $\mathbf{Q}$ and the refined feature is then passed through the subsequent network. In the decoder, after computing the pixel-wise reconstruction error $(\mathbf{X}' - \mathbf{X})^2$, SAM is then used as a weight to generate Spatial Active Reconstruction Error ($SARE$) according to Eq. (12). $SARE$ is used as the reconstruction loss to train the decoder. As shown in Figure 3, the reconstruction error caused by the background or unrelated pixels is subtracted in $SARE$ letting the APPE focus more on production-related pixels. Moreover, the correlation between SAM and progress classes bolsters the network's interpretability, elucidating the features instrumental in the model's decision-making process.

$$SARE = \sum_c\sum_i\sum_j(\mathbf{X}'_{c,i,j} - \mathbf{X}_{c,i,j})^2 SAM_{i,j}. \tag{12}$$

### C. Random Erase

As noted by [34], Class Activation Maps (CAM) typically highlight the most discriminative part of an object rather than its entirety, a phenomenon that might also occur with our proposed SAM. To address this issue, we introduce a random erase procedure during the training process.

We define the procedure to randomly erase a portion of the input image with a certain probability, dimensions, and aspect ratio. We randomly generate the erasure rectangle's position $(r_x, r_y)$ and dimensions $(r_w, r_h)$ using uniform distributions, constrained by the image width and height dimensions $w \times h$ and the specified scale and ratio parameters:

$$r_x \sim U(0, w), r_y \sim U(0, h), \tag{13}$$

$$a \sim U(a_l, a_u), b \sim U(b_l, b_u), \tag{14}$$

$$r_w = \sqrt{\frac{awh}{b}}, r_h = \sqrt{abwh}. \tag{15}$$

In this paper, we let $a_l = 0.02, a_u = 0.33$, $b_l = 0.3, b_u = 3.3$. And the erased area will be filled with black pixels, as visualized in Figure 3. Because in the training procedure, the most discriminative can be erased. By erasing the image randomly, the APPE is forced to learn not only the most discriminative part but also all the feature that may contribute to the determination of process stage.

### D. The Overall Learning Objective

Our model leverages an encoder-decoder architecture to transform image features into a GMM representation. As detailed in Sections III-B and III-A, the model's loss function comprises a weighted reconstruction loss ($SARE$), $NLL$ for fitting the Gaussian mixture distribution, and a cross-entropy loss together with a triplet loss that allows the Multi-Layer Perceptron (MLP) to learn attention weights in a self-supervised manner.

The composite loss function is summarized in Eq. (16), where $\lambda_1$, $\lambda_2$, and $\lambda_3 \in \mathbb{R}$ serve as balancing terms for the respective loss components:

$$Loss = Loss_{CE} + \lambda_1 Loss_{trip} + \lambda_2 NLL + \lambda_3 \ SARE. \quad (16)$$

Regarding the update of the parameters of the GMM for the embedded feature distribution, specifically $F_\mu(k)$ and $F_\Sigma(k)$ in Eq.(4) and Eq.(5), during the training process: While certain studies advocate for the joint training of these parameters to boost performance [26], our observations indicate that the substantial variability inherent in the normal data distribution precipitates instability during the joint training procedure. Consequently, our methodology employs an Expectation-Maximization (EM) based iterative computational approach. In every training epoch, we refine $F_\mu(k)$ and $F_\Sigma(k)$ using the maximum likelihood estimation derived from the features $\mathbf{z}$.

### E. Online Prediction and Post-Processing

Our approach, APPE, maps the image feature $\mathbf{X}_t$, generated during the production process at time $t$, to a GMM latent variable $\mathbf{z}$, which follows a GMM distribution:

$$p(\mathbf{z}) = \sum_{y=0}^{y=n-1} \alpha_y \mathcal{N}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y), \quad (17)$$

where $\alpha_y = k_{y+1} - k_y$ denotes the weight of the Gaussian component for process stage $y$, inferred from the average time cost of stage $y$ in training videos. The computed $p(\mathbf{z})$ signifies the likelihood of observing a feature value during normal assembly conditions. Thus, $p(\mathbf{z})$ is an anomaly score index, where an image with little $p(\mathbf{z})$ should be treated as an anomaly. We abstain from employing the reconstruction error as an anomaly score in this context, as the dynamic background inherent in the production process can result in challenges in accurately reconstructing even normal process images. Additionally, the marginal differences between anomalous and normal images mean that anomalous images may only exhibit minimal reconstruction loss. This phenomenon is further elucidated by the performance of AE+SARE, as detailed in Section IV.

*1) Integrating Anomaly Detection with Progress Prediction:* Building upon the aforementioned strategies, we introduce SAM and a progress-related GMM to direct the APPE's focus towards production-relevant features, thereby significantly improving anomaly detection performance. For the second component of production monitoring—progress prediction—we propose leveraging anomaly detection outcomes to refine progress predictions.

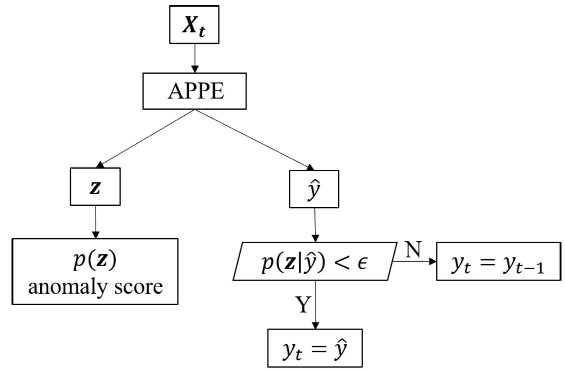


Fig. 6. Pipeline of integrating anomaly detection and progress for online prediction.

Given a progress prediction $\hat{y}$, the reliability of this prediction is quantified by the posterior probability given $\mathbf{z}$ generated in production stage $y$: $p(\mathbf{z}|\hat{y})$. Predictions associated with low confidence threshold $p(\mathbf{z}|\hat{y}) < \epsilon$, $\epsilon \in \mathbb{R}$, should be dropped and wait for a reliable prediction afterward. The online predicting and post-processing pipeline is summarized in Figure 6. At time $t$, we get an image $\mathbf{X}_t$ and compute the anomaly score and progress prediction result through the proposed APPE in an online manner. If a progress prediction is unreliable ($p(\mathbf{z}|\hat{y}) < \epsilon$), then the progress prediction for this image will be dropped, and the predicted progress will be the same as the last reliable progress prediction $y_{t-1}$.

## IV. Experimental Study

We evaluate the proposed APPE method by using two distinct datasets. The first dataset encompasses a water valve production process, where the assembled parts are the predominant focus of the image data. The second dataset represents a real-world production scenario, encompassing not only assembly operations but also tasks such as lamination, coating, and vacuum extraction, involving both mechanical parts and flexible composite materials.

Our project's testing environment is configured with Python 3.8 and CentOS 7.9. To ensure optimal performance and efficiency in our training process, we utilized a setup comprising either a single A100 GPU or a dual A100 GPU. To maintain fairness in our evaluations, all models were standardized using the ResNet50 architecture as their backbone [40].

### A. Water Valve Production

In this section, we present the performance of our proposed model on a dataset from a single-station water valve production process. The production stages is visualized in Figure 7, including: (1) assembling the intermediate valve body; (2) fastening the screw; (4) assembling the middle part of the valve; (5) assembling the upper part of the valve; (6) assembling the rocker arm; (7) assembling parts onto the rocker arm; (8) completing the lower clamp assembly of the water cut-off valve; and finally, (9) assembling the upper clamp. The whole process lasts for about 3000 time-frames per cycle.

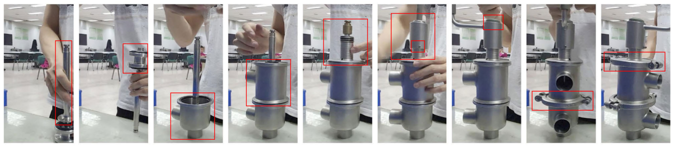

Fig. 7. Visualized production stages.

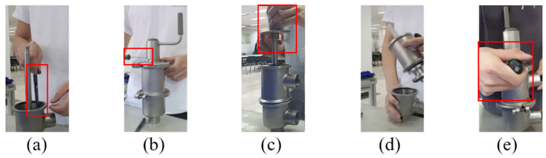


Fig. 8. Examples of anomalies in assembly.

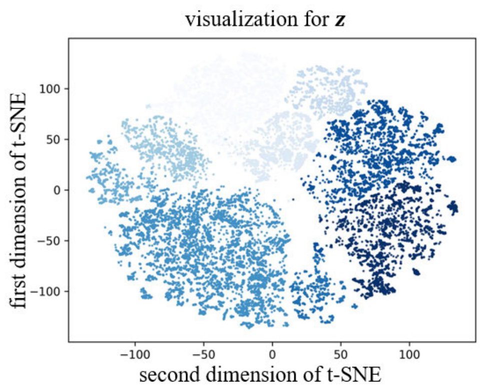


Fig. 9. t-SNE Visualization of features **z** of Water valve production dataset.

The training set contains 11 assembly videos and is extracted as 29529 images for training. The test set contains 31 assembly videos and is annotated at ten time frames per interval as normal or anomalous, with a total of 700 anomaly data points and 5790 normal data points. Anomalies in the dataset, as shown in Figure 8, include foreign object anomalies (a), worker operation anomalies (d,e), progress correlated anomalies (b), and part interaction anomalies (d,c). It is important to note that some anomaly images may exhibit characteristics of multiple cases of anomaly and our focus is detecting if there exists any anomaly no matter what kind it is.

We utilize t-distributed Stochastic Neighbor Embedding (t-SNE) to visualize the high-dimensional feature space for encoded $\mathbf{z}$. t-SNE is a powerful dimensionality reduction technique that helps to project high-dimensional data into two or three dimensions while preserving the relative distances between data points. The visualization of the model's features is shown in Figure 9. The gradient from light to dark shades corresponds to the process stages of the image within the video sequence, with darker shades representing later stages in the assembly process. We can see that each process class is concentrated together either for the training set or testing set. The visualized SAM and reconstructed $\mathbf{X}'$ are shown in Figure 10. This visualized result shows that SAM mainly focuses on the location where parts being assembled, will be assembled or hands performing production-related operations.

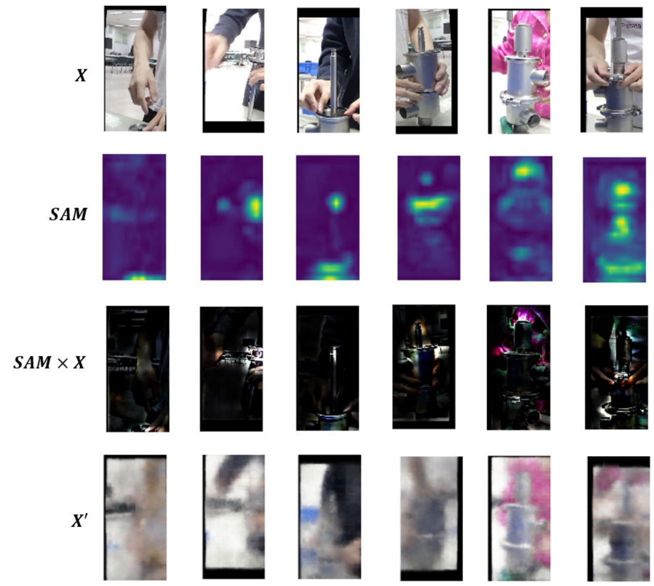


Fig. 10. Visualized SAM and reconstructed $\mathbf{X}'$.

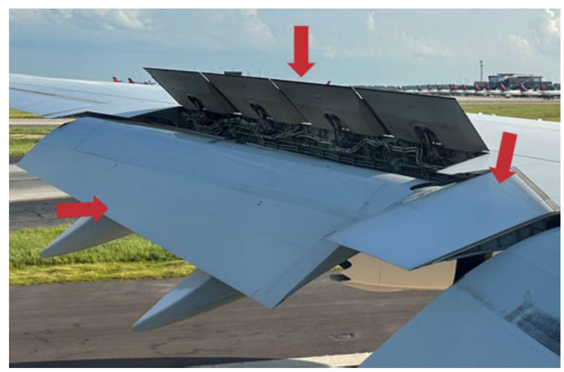

Fig. 11. The spoilers of aircraft.

## B. Commercial Aircraft Spoiler Production

In this study, we evaluate our model's performance in the real-world production of commercial aircraft spoilers. This workstation is selected for its involvement with both mechanical parts and flexible materials. The finished spoiler product is depicted in Figure 11. The production process is multifaceted, involving initial pre-assembly work next to tooling, followed by steps including primer drying, inspection of auxiliary materials, layering of composite materials, and applying films of various colors. Subsequent stages involve assembling the rigid structure, covering it with isolation film, and additional steps like bagging and leak testing, culminating in autoclave curing. This paper focuses on the spoiler production stages up to the autoclave process, a period of approximately 3 hours characterized by mixed-line and entirely manual manufacturing. One particular stage, involving waiting and inspecting, shows no visible changes in the workpiece's appearance. Additionally, the tooling's position varies in each cycle within the camera's shooting range. Visual image data was collected using the workshop's monitoring cameras, ensuring no disruption to production or additional hardware costs.

Our dataset is meticulously compiled from three weeks of factory surveillance footage. By extracting one frame

TABLE I
DATASET DESCRIPTION FOR COMMERCIAL AIRCRAFT SPOILER PRODUCTION

| Training | video 1 | video 2 | video 3 | video 4 | video 5 | video 6 |
|---|---|---|---|---|---|---|
| # of images | 3063 | 2256 | 2844 | 2063 | 2613 | 2099 |
| type of spoiler | D | B | C | C | B | C |

| Testing | video 1 | video 2 | video 3 | video 4 | video 5 | video 6 | video 7 |
|---|---|---|---|---|---|---|---|
| # of images | 2113 | 5481 | 2862 | 1695 | 2442 | 2532 | 1956 |
| type of spoiler | A | A | B | B | C | B | C |

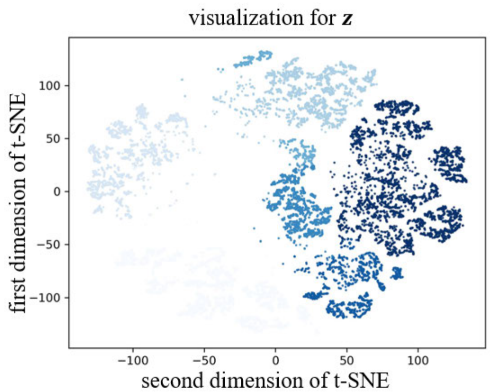


Fig. 12. t-SNE Visualization of features **z** of spoiler production dataset.

every three seconds, we amass a comprehensive collection of approximately 46,000 images. This dataset is methodically divided into distinct units: six for training, and seven for testing, as elaborated in Table I. The training set includs footage of three different spoiler types, while the test and validation sets comprise spoilers from both the training set and an additional, previously untrained spoiler type. Variations in spoilers are evident in their shapes (from rectangular to trapezoidal), sizes, and specific structures for wing attachment. Despite these variations, the production process remains consistent across different spoiler types. Production data during factory downtime is excluded based on the operational hours of the facility. For the evaluation of anomaly detection, the test data is labeled with the collective expertise of multiple factory personnel and managerial oversight. In total, 23 distinct anomaly circumstances and 1847 normal circumstances are labeled. Among the anomalies are instances of pre-assembly parts getting jammed, workers remaining seated at the tooling stations during active assembly phases, and premature relocation of tooling and equipment before the completion of the assembly cycle. The visualized features are shown in Figure 12 with the same interpretation as Figure 9.

### *C. Baseline Comparison*

Here we compare our method with three well-known anomaly detection algorithms: an AE [41] with reconstruction error as anomaly score; VAE [25] wherein the reconstruction probability serves as the anomaly indicator and DaGMM [26]. In addition, we also test the performance of using SARE in Eq. 12 instead of AE's reconstruction loss (the column AE+SARE in the table), to prove the effectiveness of SAM in focusing on production related feature.

In our study, we employ two widely recognized metrics: the Area Under the receiver operating characteristic Curve (AUC) and the Equal Error Rate (EER), to evaluate the effectiveness of various methods. The quantitative results, as detailed in Table II and III, demonstrate that our proposed method surpasses all baseline models in overall anomaly detection performance. Notably, we observ that the AE augmented with the SAM (AE+SARE) outperform the other three baseline models. This indicates that SAM's role in directing the model's focus towards production-relevant features is indeed effective. However, the detection performance still is not satisfactory compared with or proposed APPE. The primary reason is that although SAM highlights production-related features, normal images are still challenging to reconstruct. This is largely because, in typical production settings, the number of training samples available is often limited and may not cover all normal conditions present in the test set. As a result, the model's decoder part struggles to learn effectively, leading to significant reconstruction loss for normal images (as shown in the last row of Figure 10). However, the encoder part, which only needs to extract relevant information for reconstructing production-related features from images, faces a much simpler task. Therefore, analyzing feature values' probabilities instead of the reconstruction loss can circumvent the issues associated with a poorly trained decoder.

Furthermore, our model exhibits substantial superiority over all baseline models in detecting specific anomaly subclasses, including foreign object, worker operation, part interaction, and progress-correlated anomalies, as shown in Table II.

### *D. Ablation Studies*

To rigorously assess the indispensability of each component within the APPE framework, we conduct a series of ablation studies. These studies involve variations of the APPE model; including the substitution of SAM with CAM, denoted as $\text{APPE}_{CAM}$; the removal of SAM, indicated as APPE-noSAM; the exclusion of $NLL$, referred to as APPE-noNLL; and the exclusion of random erase, referred to as APPE-noErase. Through these modifications, we aim to elucidate the contribution of each individual element to the overall performance of the APPE model.

The results from our ablation studies are presented in Table IV. As indicated in the second column, it is evident that our SAM outperforms alternatives that employ CAM as a substitute. This superiority stems from CAM's limited focus on features that determine the current stage of progress, whereas SAM enables the capture of all production-relevant features. This approach not only prevents the loss of crucial information but also filters out irrelevant production features.

TABLE II
QUANTITATIVE RESULT FOR ANOMALY DETECTION IN WATER VALVE PRODUCTION

| Anomaly | | Proposed | AE | AE+SARE | VAE | DaGMM |
|---|---|---|---|---|---|---|
| Foreign object | AUC | **88.71** $\pm$ 3.03 | 66.54 $\pm$ 2.09 | 72.46 $\pm$ 6.12 | 59.12 $\pm$ 4.37 | 55.80 $\pm$ 3.71 |
| | EER | **19.57** $\pm$ 3.05 | 40.71 $\pm$ 1.76 | 36.08 $\pm$ 5.17 | 42.83 $\pm$ 2.94 | 44.71 $\pm$ 2.94 |
| Worker operation | AUC | **88.7** $\pm$ 1.97 | 53.67 $\pm$ 1.61 | 56.56 $\pm$ 4.20 | 56.56 $\pm$ 1.06 | 51.75 $\pm$ 2.49 |
| | EER | **18.19** $\pm$ 1.09 | 49.49 $\pm$ 2.89 | 46.79 $\pm$ 4.37 | 45.15 $\pm$ 2.93 | 48.79 $\pm$ 2.12 |
| Part interaction | AUC | **94.8** $\pm$ 0.46 | 43.09 $\pm$ 0.77 | 71.14 $\pm$ 2.21 | 39.87 $\pm$ 1.80 | 55.89 $\pm$ 2.70 |
| | EER | **11.32** $\pm$ 0.11 | 56.85 $\pm$ 0.61 | 34.56 $\pm$ 2.63 | 57.86 $\pm$ 2.59 | 45.71 $\pm$ 1.71 |
| Progress correlated | AUC | **84.39** $\pm$ 1.89 | 61.69 $\pm$ 1.13 | 63.98 $\pm$ 5.07 | 63.33 $\pm$ 1.06 | 49.56 $\pm$ 3.14 |
| | EER | **24.27** $\pm$ 2.9 | 42.28 $\pm$ 1.30 | 41.17 $\pm$ 4.56 | 39.54 $\pm$ 1.28 | 49.23 $\pm$ 3.24 |
| All | AUC | **90.79** $\pm$ 0.7 | 54.64 $\pm$ 0.79 | 65.87 $\pm$ 3.18 | 50.17 $\pm$ 1.22 | 54.17 $\pm$ 1.96 |
| | EER | **17.49** $\pm$ 0.98 | 48.44 $\pm$ 0.81 | 39.39 $\pm$ 2.53 | 50.75 $\pm$ 1.85 | 46.67 $\pm$ 1.48 |

TABLE III
QUANTITATIVE RESULT FOR ANOMALY DETECTION IN COMMERCIAL AIRCRAFT SPOILER PRODUCTION

| | Proposed | AE | AE+SARE | VAE | DaGMM |
|---|---|---|---|---|---|
| AUC | **80.45** $\pm$ 6.39 | 40.22 $\pm$ 7.93 | 50.37 $\pm$ 2.95 | 41.79 $\pm$ 0.94 | 52.51 $\pm$ 10.93 |
| EER | **27.24** $\pm$ 4.99 | 57.05 $\pm$ 8.27 | 51.63 $\pm$ 5.46 | 57.24 $\pm$ 1.78 | 48.41 $\pm$ 5.97 |

TABLE IV
QUANTITATIVE RESULT FOR ABLATION STUDY

| Anomaly | | Proposed (APPE) | APPE$_{CAM}$ | APPE-noSAM | APPE-noNLL | APPE-noErase |
|---|---|---|---|---|---|---|
| Foreign object | AUC | **88.71** $\pm$ 3.03 | 76.32 $\pm$ 2.39 | 50.6 $\pm$1.64 | 85.34 $\pm$ 1.39 | 83.19$\pm$2.31 |
| | EER | **19.57** $\pm$ 3.05 | 28.42 $\pm$ 1.73 | 51.53 $\pm$ 1.01 | 21.82 $\pm$ 1.01 | 23.71 $\pm$ 2.07 |
| Worker operation | AUC | **88.7** $\pm$ 1.97 | 69.24 $\pm$ 2.58 | 62.1$\pm$ 0.77 | 85.63 $\pm$ 1.81 | 86.50 $\pm$ 2.79 |
| | EER | **18.19** $\pm$ 1.09 | 38.44 $\pm$ 3.35 | 43.02$\pm$ 1.69 | 21.73 $\pm$ 2.62 | 20.07 $\pm$ 3.54 |
| Part interaction | AUC | **94.8**$\pm$ 0.46 | 67.26 $\pm$ 1.84 | 51.17 $\pm$ 1.53 | 94.36 $\pm$ 0.28 | 93.75 $\pm$ 1.19 |
| | EER | **11.32** $\pm$ 0.11 | 38.87 $\pm$ 1.81 | 51.75 $\pm$ 1.53 | 11.63 $\pm$ 0.4 | 13.59 $\pm$ 1.48 |
| Progress correlated | AUC | 84.39 $\pm$ 1.89 | 64.09 $\pm$ 2.0 | 57.48$\pm$ 0.71 | **87.38** $\pm$ 2.06 | 76.57 $\pm$ 2.62 |
| | EER | 24.27$\pm$ 2.9 | 40.89 $\pm$ 1.68 | 45.76 $\pm$ 1.53 | **20.04** $\pm$ 2.51 | 28.75 $\pm$ 2.24 |
| **All** | AUC | **90.79** $\pm$0.7 | 67.79 $\pm$ 0.82 | 54.23 $\pm$ 0.69 | 89.55 $\pm$ 0.3 | 87.38 $\pm$ 0.64 |
| | EER | **17.49** $\pm$ 0.98 | 38.51 $\pm$ 0.62 | 49.42 $\pm$ 1.03 | 18.64 $\pm$ 0.67 | 20.56 $\pm$ 1.12 |

TABLE V
ANOMALY DETECTION WITH DIFFERENT NUMBERS OF ANNOTATED PRODUCTION STAGES

| Anomaly | | Proposed (n=9) | n=8 | n=6 | n=4 | n=3 | n=2 |
|---|---|---|---|---|---|---|---|
| Foreign object | AUC | **88.71** | 83.69 | 84.72 | 84.58 | 65.09 | 45.88 |
| | EER | **19.57** | 22.22 | 23.93 | 22.68 | 40.08 | 52.17 |
| Worker operation | AUC | **88.7** | 86.20 | 82.93 | 64.95 | 76.35 | 77.88 |
| | EER | **18.19** | 20.19 | 25.35 | 38.43 | 28.20 | 29.10 |
| Part interaction | AUC | **94.8** | 92.84 | 91.63 | 89.41 | 79.43 | 61.27 |
| | EER | **11.32** | 14.12 | 14.68 | 17.27 | 24.45 | 42.06 |
| Progress correlated | AUC | **84.39** | 83.13 | 83.03 | 75.51 | 68.39 | 54.98 |
| | EER | **24.27** | 25.04 | 24.59 | 30.36 | 36.03 | 47.12 |
| All | AUC | **90.79** | 87.73 | 86.53 | 82.39 | 73.63 | 61.80 |
| | EER | **17.49** | 19.75 | 22.08 | 24.85 | 31.53 | 41.95 |

Additionally, the overall performance of both APPE-noNLL and APPE-noSAM, as shown in the third and fourth columns, is inferior to that of our proposed APPE model, thereby underscoring the indispensable nature of both SAM and the progress-related GMM embedding proposed in this study. Besides, we observer that the subtraction of NLL influences much less compared with the subtraction of SAM. This is because after the subtraction of NLL, the model become a metrics learning without considering the correlation matrix of features vectors. And information loss from the correlation matrix is minor compared with the progress related attention SAM which is the key in our anomaly detection. Furthermore, APPE outperforms APPE-noErase, proofing the proposed random erase procedure in III-C being vital in improving the proposed APPE performance.

### *E. Sensitivity Analysis*

Although we can apply the method proposed in [37] to optimally and unsupervisedly divide and annotate process stages, it is also important to explore the impact of suboptimal progress information on anomaly detection. To explore this issue, we conduct a sensitivity analysis by varying the number of annotated production stages. We investigate how the anomaly detection performance changes as the number of annotated production stages $n$ decreases. The quantitative results are shown in Table V and Figure 13.

As shown in Figure 13, we observe that as the number of annotated progress stages decreases, the detection performance for different types of anomalies declines. However, we also notice some interesting phenomena: when the number of stages $n$ is greater than 4, the model's anomaly detection per-

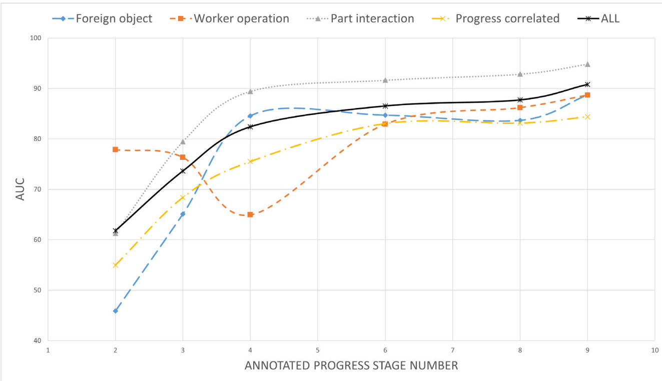


Fig. 13. Sensitivity analysis for anomaly detection with different number of annotated production stages on water valve production dataset.

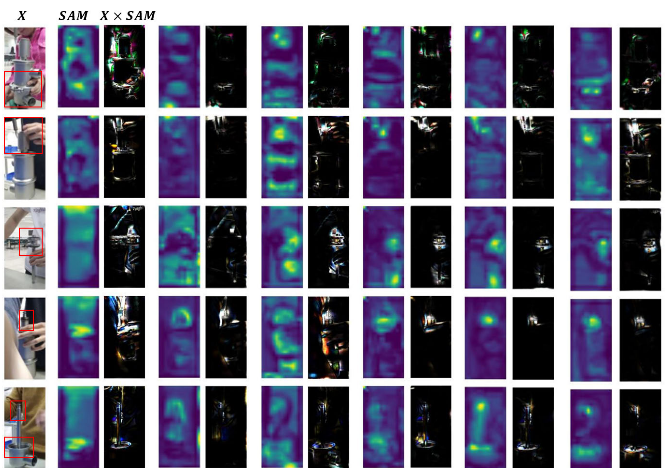


Fig. 14. SAM learned under different numbers of annotated production stages.

formance does not significantly drop. But when $n$ is less than or equal to 4, the model's performance decreases significantly as the number of annotated stages decreases. To explore the underlying mechanism behind this phenomenon, we visualize the SAM learned by the model under different $n$ in Figure 14. In the first column of the figure, the red box indicates the part of the workpiece currently being operated on, which needs to be focused on. We can see that when the number of stages decreases to 4 or less, the attention weights learned by SAM are no longer accurate, and the model cannot focus on the part being operated on. In some images with $n = 4$, SAM even completely focuses on incorrect positions, which can be the cause of detection drop of worker operation anomaly when $n = 4$. The above results indicate that the decrease of $n$ makes SAM unable to learn accurate attention weights which leads to detection performance drop of anomaly detection.

### F. Integrating Anomaly Detection With Progress Prediction

In addition to anomaly detection, our model concurrently outputs predictions for the current stage of the production process, denoted as $\hat{y}$. Anomalous operations can adversely affect the accuracy of stage recognition. Consequently, as discussed in Section III-E, we employ post-processing on the recognition results based on the probability of the ongoing process stage $\hat{y}$ to get the final prediction $y_t$ for time $t$. We select a video depicting normal operations as the validation set and compute the 99% quantile of $p(\mathbf{z}|\hat{y})$ to determine the threshold value $\epsilon$. The remaining 30 test videos are used as the test set for evaluating the performance of progress prediction.

TABLE VI
PROGRESS PREDICTION ACCURACY BEFORE AND AFTER INTEGRATING WITH ANOMALY DETECTION

| Data sets | Situation | Metric | After | Before (MLP) |
|---|---|---|---|---|
| Airplane spoiler production | All | Acc | **92.36**± 4.30 | 92.25± 4.24 |
| | | MAPE | **4.95**±0.45 | 4.97± 0.44 |
| | Only anomaly | Acc | **86.28**± 5.49 | 85.142± 4.69 |
| | | MAPE | **7.13**±0.31 | 7.24± 0.16 |
| Water valve production | All | Acc | **95.37**± 0.60 | 94.60± 0.48 |
| | | MAPE | **4.97**±0.08 | 5.03± 0.06 |
| | Only anomaly | Acc | **84.11**± 3.49 | 74.18± 3.74 |
| | | MAPE | **5.80**±0.43 | 7.77± 0.78 |

To assess the efficacy of our model in predicting the progression of tasks, we utilize two key metrics. The primary metric is classification accuracy, which directly measures the model's ability to correctly identify the current stage of the production process. The second metric, Mean Absolute Percentage Error (MAPE) of time, is calculated as per Eq. 18:

$$MAPE\% = \frac{1}{N}\sum_{i=0}^{N-1}\sum_{t=0}^{T_{i,n}-1}\frac{\left|t-\frac{T_{i,(y_{i,t})+1}+T_{i,y_{i,t}}}{2}\right|}{T_{i,n}}, \tag{18}$$

where $T_{i,y_{i,t}}$ denotes the starting time of the predicted process stage $y_{i,t}$ in video $i$ at time $t$, $T_{i,n}$ is the ending time of the last stage (i.e., the total number of frames in video $i$), and $N$ is the number of videos in the testing set. This formulation portrays progress prediction as a continuous endeavor by computing the temporal midpoint, $\frac{T_{i,y_{i,t}}+T_{i,(y_{i,t})+1}}{2}$, of the anticipated category as the predicted value. This approach not only evaluates the likelihood of accurate model predictions but also quantitatively measures the model's prediction deviation by juxtaposing this predicted value with the actual occurrence time. This dual assessment strategy effectively captures both the precision in distinguishing distinct process stages and the accuracy of temporal predictions.

The comparative results before and after integration are presented in Table VI. It is observed that both accuracy and Mean Absolute Percentage Error (MAPE) improve across both datasets following integration, especially under anomaly situations. This demonstrates that integrating anomaly detection with progress prediction can enhance the performance of both tasks.

### G. Discussion

In conclusion, the comparative analyses with baseline models across two distinct datasets, as illustrated in Table II and Table III, unequivocally demonstrates the superior efficacy of our proposed APPE approach in detecting all the anomalies summarized in Figure 2. Further reinforcement of our model's effectiveness is provided by the ablation studies detailed in Table IV, which highlight the critical contribution of each component within our framework. Our APPE model adeptly addresses the dual challenges of anomaly detection and progress prediction within a singular, cohesive framework. This integration not only halves the computational complexity traditionally associated with handling these tasks independently but also synergistically enhances the performance of both functions. Moreover, the visual insights offered by the SAM, exemplified in Figure 10.

## V. Conclusion

In this study, we propose the APPE model, designed to map images from normal production operations to a GMM, taking into account the heterogeneity of image features across different stages of the process. The implementation of the SAM as an attention mechanism within both the encoder and decoder significantly enhances the APPE's focus on production-relevant features. Furthermore, SAM contributes to improving the interpretability of the APPE model, allowing for a clearer understanding of how it identifies and reacts to various features within the production environment.

Our approach uniquely integrates progress prediction and anomaly detection into a cohesive task, leveraging the SAM and progress-related GMM to establish a framework enhancing the anomaly detection performance. This also enables the refinement of progress prediction through posterior anomaly scoring, resulting in improved accuracy and MAPE across both datasets examined.

Through the lens of two case studies, the APPE model demonstrates superior performance in anomaly detection, achieving higher AUC and EER metrics compared to four well-established baselines. This underscores the effectiveness and adaptability of our model in addressing the complexities of real-world production monitoring, marking a significant advancement in the fields of anomaly detection and progress prediction within industrial environments.

For future research, exploring anomaly clustering to classify various types and diagnose the causes of anomalies presents a compelling direction. Additionally, considering the use of multi-frame images as input to account for the temporal association of features offers another avenue for advancing the field.

## VI. Data and Code Availability

To uphold the principles of transparency and reproducibility in research, we provide the code, model checkpoints, logs for the method proposed in this paper with this link: https://cloud.tsinghua.edu.cn/d/cfb8aeafcd174b748167/. And the dataset for the water-valve production can be found at [37].

## References


[1] L. Zhou, H. Wang, C. Berry, X. Weng, and S. J. Hu, "Functional morphing in multistage manufacturing and its applications in high-definition metrology-based process control," *IEEE Trans. Autom. Sci. Eng.*, vol. 9, no. 1, pp. 124–136, Jan. 2012.

[2] D. Li, J. Lu, T. Zhang, and J. Ding, "Self-supervised learning and multisource heterogeneous information fusion based quality anomaly detection for heavy-plate shape," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 2, pp. 1223–1234, Apr. 2024.

[3] R. Guo, K. Guo, and J. Dong, "Phase partition and online monitoring for batch process based on multiway BEAM," *IEEE Trans. Autom. Sci. Eng.*, vol. 14, no. 4, pp. 1582–1589, Oct. 2017.

[4] B. Ramachandra, M. J. Jones, and R. R. Vatsavai, "A survey of single-scene video anomaly detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2293–2312, May 2022.

[5] V. Zavrtanik, M. Kristan, and D. Skočaj, "DRAEM—A discriminatively trained reconstruction embedding for surface anomaly detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8330–8339.

[6] H. Yan, M. Grasso, K. Paynabar, and B. M. Colosimo, "Real-time detection of clustered events in video-imaging data with applications to additive manufacturing," *IISE Trans.*, vol. 54, no. 5, pp. 464–480, 2021, doi: 10.1080/24725854.2021.1882013.

[7] R. J. Kuo and F. F. Nursyahid, "Foreign objects detection using deep learning techniques for graphic card assembly line," *J. Intell. Manuf.*, vol. 34, no. 7, pp. 2989–3000, Oct. 2023, doi: 10.1007/s10845-022-01980-7.

[8] C. Chen et al., "Monitoring of assembly process using deep learning technology," *Sensors*, vol. 20, no. 15, p. 4208, Jul. 2020.

[9] J. Wang, C. Chen, and C. Dai, "A mechanical assembly monitoring method based on domain adaptive semantic segmentation," *Int. J. Adv. Manuf. Technol.*, vol. 128, nos. 1–2, pp. 625–637, Sep. 2023.

[10] C. Chen, C. Zhang, C. Li, and J. Hong, "Assembly monitoring using semantic segmentation network based on multiscale feature maps and trainable guided filter," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.

[11] H. Huang, Z. Wei, and L. Yao, "A novel approach to component assembly inspection based on mask R-CNN and support vector machines," *Information*, vol. 10, no. 9, p. 282, Sep. 2019.

[12] H. Ben Abdallah, I. Jovančević, J.-J. Orteu, and L. Brèthes, "Automatic inspection of aeronautical mechanical assemblies by matching the 3D CAD model and real 2D images," *J. Imag.*, vol. 5, no. 10, p. 81, Oct. 2019.

[13] M. Kim et al., "A vision-based system for monitoring block assembly in shipbuilding," *Comput.-Aided Des.*, vol. 59, pp. 98–108, Feb. 2015.

[14] W. Tao, M. Al-Amin, H. Chen, M. C. Leu, Z. Yin, and R. Qin, "Real-time assembly operation recognition with fog computing and transfer learning for human-centered intelligent manufacturing," *Proc. Manuf.*, vol. 48, pp. 926–931, Jan. 2020.

[15] C. Chen, C. Li, D. Li, Z. Zhao, and J. Hong, "Mechanical assembly monitoring method based on depth image multiview change detection," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021.

[16] J. Patalas-Maliszewska, D. Halikowski, and R. Damaševičius, "An automated recognition of work activity in industrial manufacturing using convolutional neural networks," *Electronics*, vol. 10, no. 23, p. 2946, Nov. 2021.

[17] C. Chen, T. Wang, D. Li, and J. Hong, "Repetitive assembly action recognition based on object detection and pose estimation," *J. Manuf. Syst.*, vol. 55, pp. 325–333, Apr. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0278612520300625

[18] L. Liu, Y. Liu, and J. Zhang, "Learning-based hand motion capture and understanding in assembly process," *IEEE Trans. Ind. Electron.*, vol. 66, no. 12, pp. 9703–9712, Dec. 2019.

[19] M. Ai, Y. Xie, S. X. Ding, Z. Tang, and W. Gui, "Domain knowledge distillation and supervised contrastive learning for industrial process monitoring," *IEEE Trans. Ind. Electron.*, vol. 70, no. 9, pp. 9452–9462, Sep. 2023.

[20] Y. Lyu, J. Chen, and Z. Song, "Image-based process monitoring using deep learning framework," *Chemometric Intell. Lab. Syst.*, vol. 189, pp. 8–17, Jun. 2019.

[21] R. Xu, J. Wu, X. Yue, and Y. Li, "Online structural change-point detection of high-dimensional streaming data via dynamic sparse subspace learning," *Technometrics*, vol. 65, no. 1, pp. 19–32, Jan. 2023, doi: 10.1080/00401706.2022.2046171.

[22] S. Suriano, H. Wang, C. Shao, S. J. Hu, and P. Sekhar, "Progressive measurement and monitoring for multi-resolution data in surface manufacturing considering spatial and cross correlations," *IIE Trans.*, vol. 47, no. 10, pp. 1033–1052, Oct. 2015.

[23] C. Liu, Z. Kong, S. Babu, C. Joslin, and J. Ferguson, "An integrated manifold learning approach for high-dimensional data feature extractions and its applications to online process monitoring of additive manufacturing," *IISE Trans.*, vol. 53, no. 11, pp. 1215–1230, 2021, doi: 10.1080/24725854.2020.1849876.

[24] A. AlBahar, I. Kim, and X. Yue, "A robust asymmetric kernel function for Bayesian optimization, with application to image defect detection in manufacturing systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 4, pp. 3222–3233, Oct. 2022.

[25] D. Park, Y. Hoshi, and C. C. Kemp, "A multimodal anomaly detector for robot-assisted feeding using an LSTM-based variational autoencoder," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 1544–1551, Jul. 2018.

[26] B. Zong et al., "Deep autoencoding Gaussian mixture model for unsupervised anomaly detection," in *Proc. Int. Conf. Learn. Represent.*, 2018. [Online]. Available: https://openreview.net/pdf?id=BJJLHbb0-
[27] M. Sabokrou, M. Fayyaz, M. Fathy, and R. Klette, "Deep-cascade: Cascading 3D deep neural networks for fast anomaly detection and localization in crowded scenes," *IEEE Trans. Image Process.*, vol. 26, no. 4, pp. 1992–2004, Apr. 2017.
[28] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023. [Online]. Available: https://www.ncbi.nlm.nih.gov/pubmed/35180075
[29] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," 2014, *arXiv:1409.0473*.
[30] S. Liu, Y. Chen, K. Liu, and J. Zhao, "Exploiting argument information to improve event detection via supervised attention mechanisms," in *Proc. 55th Annu. Meeting Assoc. Comput. Linguistics*, Jan. 2017, pp. 1789–1798.
[31] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.
[32] M. Meng, T. Zhang, Q. Tian, Y. Zhang, and F. Wu, "Foreground activation maps for weakly supervised object localization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3385–3395.
[33] J. Xu et al., "CREAM: Weakly supervised object localization via class RE-activation mapping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9437–9446.
[34] L. Zhu et al., "Background-aware classification activation map for weakly supervised object localization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14175–14191, Dec. 2023. [Online]. Available: https://www.ncbi.nlm.nih.gov/pubmed/37643092
[35] R. Xu, C. Wang, S. Xu, W. Meng, and X. Zhang, "Wave-like class activation map with representation fusion for weakly-supervised semantic segmentation," *IEEE Trans. Multimedia*, vol. 26, pp. 581–592, 2023.
[36] L. Huang, L. Wang, and H. Li, "Foreground-action consistency network for weakly supervised temporal action localization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8002–8011.
[37] Y. Li, Z.-H. Zhang, X. Yue, and L. Zheng, "An unsupervised embedding method based on streaming videos for process monitoring in repetitive production systems," *IISE Trans.*, pp. 1–16, Sep. 2024, doi: 10.1080/24725854.2024.2386415.
[38] Z. Jiang, Y. Zheng, H. Tan, B. Tang, and H. Zhou, "Variational deep embedding: An unsupervised and generative approach to clustering," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, Aug. 2017, pp. 1965–1972.
[39] L. Yang, W. Fan, and N. Bouguila, "Clustering analysis via deep generative models with mixture models," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 1, pp. 340–350, Jan. 2022. [Online]. Available: https://www.ncbi.nlm.nih.gov/pubmed/33048769
[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[41] M. Maggipinto, A. Beghi, and G. A. Susto, "A deep convolutional autoencoder-based approach for anomaly detection with industrial, non-images, 2-dimensional data: A semiconductor manufacturing case study," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 3, pp. 1477–1490, Jul. 2022.

**Yifan Li** received the B.S. degree from the Department of Industrial Engineering, Tsinghua University, Beijing, China, and the B.S. degree from the Department of Mechanical Engineering, Tsinghua University, in 2021. He is currently pursuing the Ph.D. degree with the Department of Industrial Engineering, Tsinghua University.

He was a Visiting Student at UC Berkeley in 2020. He is a Visiting Ph.D. Scholar at ISyE, Georgia Institute of Technology. His research interests focus on machine learning for advanced manufacturing, learning to optimize, computer vision, medical image, and large vision-language model. His researches have received several awards, including the Winner of the IISE QCRE Best Student Paper Award and the Winner of the INFORMS QSR Best Student Paper Award.

**Zhi-Hai Zhang** (Senior Member, IEEE) received the B.S. and Ph.D. degrees in mechanical engineering from Tsinghua University in 1997 and 2002, respectively. He is currently an Associate Professor with the Industrial Engineering Department, Tsinghua University. His current research interests include production and operational management, resource allocation optimization, data-driven decision support, supply chain and logistics management, and production planning and scheduling. He is a member of the IEEE Robotics and Automation Society (RAS), the Institute for Operations Research and the Management Sciences (INFORMS), and the Institute of Industrial and Systems Engineers (IISE).

**Jiaqi Xu** received the dual B.S. degree in industrial engineering and economics and the Ph.D. degree in industrial engineering from Tsinghua University in 2018 and 2024, respectively.

He is currently an Assistant Research Fellow with China Waterborne Transport Research Institute, Beijing, China. His research pursuits primarily revolve around robust optimization, information-driven manufacturing, and learning to optimize.

**Xiaowei Yue** (Senior Member, IEEE) received the B.S. degree in mechanical engineering from Beijing Institute of Technology, Beijing, China, in 2011, the M.S. degree in power engineering and engineering thermophysics from Tsinghua University, Beijing, in 2013, and the M.S. degree in statistics and the Ph.D. degree in industrial engineering from Georgia Institute of Technology, Atlanta, USA, in 2016 and 2018, respectively.

Currently, he is an Associate Professor at the Department of Industrial Engineering, Tsinghua University. Prior to that, he was an Assistant Professor at the Grado Department of Industrial and Systems Engineering, Virginia Tech, Blacksburg, USA. His research interests focus on machine learning for advanced manufacturing. He is a Senior Member of ASQ and IISE and a member of ASME and SME. He was a recipient of the IISE Hamed K. Eldin Outstanding Early Career IE in Academia Award, the SME Outstanding Young Manufacturing Engineer Award, the IISE Manufacturing and Design Young Investigator Award, more than ten best paper awards, and two best dissertation awards. He received the Grainger Frontiers of Engineering Grant Award from U.S. National Academy of Engineering (NAE). He serves as an Associate Editor for *IISE Transactions*, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS. He is selected to be an Editorial Board Member for *PNAS Nexus*, an open-access journal of the U.S. National Academy of Sciences (NAS).

**Li Zheng** received the B.S. and Ph.D. degrees from Tsinghua University, Beijing, China, in 1986 and 1991, respectively.

He was a Visiting Professor with Georgia Institute of Technology, Atlanta, GA, USA, from 1994 to 1996. He is currently a Professor with the Department of Industrial Engineering, Tsinghua University. He has published academic articles in various well-established journals, including *IISE Transactions*, IEEE TRANSACTIONS ON AUTOMATION SCIENCE AND ENGINEERING, *Production and Operations Management*, *European Journal of Operational Research*, and *International Journal of Production Research*. His research interests include production system analysis, vision based production monitoring, and information-driven manufacturing.

Prof. Zheng is an IISE Fellow. He received several important awards, such as the National Science and Technology Progress Award in 2005 and the National Invention Award in 1990.